

A Counterexample to Sequential-ASQ Renormability for All Spaces Containing c_0

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2026-06-20

Status

This packet gives a candidate ZFC counterexample, likely valid, to the renorming question asked in Aviles–Martinez–Cervantes–Rueda Zoca [1, p. 2]. The source paper asks whether every Banach space containing c_0 can be equivalently renormed to be sequentially ASQ. We show that the answer is no: ℓ_∞ contains c_0 , but no equivalent real norm on ℓ_∞ is sequentially ASQ.

Proof Intuition

The obstruction is that a sequentially ASQ witness carries hidden complementability. Let $(x_n) \subset S_X$ witness sequential ASQ, and choose norming functionals $f_n(x_n) = 1$. Since

$$\|x_n \pm tx\| \longrightarrow 1 \quad (0 < t\|x\| < 1),$$

the inequalities obtained by applying f_n force $f_n(x) \rightarrow 0$ for each fixed $x \in X$. After passing to a diagonal subsequence, the matrix $(f_i(x_j))$ is a small perturbation of the identity on c_0 . The map $x \mapsto (f_i(x))$ therefore gives, after inverting this matrix, a bounded projection from X onto the c_0 -copy generated by the chosen subsequence.

The Obstruction Lemma

Lemma 1. *Let X be a real sequentially ASQ Banach space, witnessed by $(x_n) \subset S_X$. For each n , choose $f_n \in S_{X^*}$ with $f_n(x_n) = 1$. Then $f_n(x) \rightarrow 0$ for every $x \in X$. Also (x_n) is weakly null.*

Proof. Fix $x \in X$. If $x = 0$ there is nothing to prove. Choose $t > 0$ with $t\|x\| < 1$. By sequential ASQ, applied to tx and to $-tx$,

$$\|x_n + tx\| \rightarrow 1, \quad \|x_n - tx\| \rightarrow 1.$$

Since $f_n(x_n) = 1$ and $\|f_n\| = 1$,

$$1 + tf_n(x) \leq \|x_n + tx\|, \quad 1 - tf_n(x) \leq \|x_n - tx\|.$$

Letting $n \rightarrow \infty$ gives $tf_n(x) \rightarrow 0$, hence $f_n(x) \rightarrow 0$.

Now let $g \in X^*$. We prove $g(x_n) \rightarrow 0$. By homogeneity assume $\|g\| = 1$. If not, for some $\delta > 0$ there is a subsequence along which $|g(x_n)| \geq \delta$. Passing to a further subsequence and replacing g

by $-g$ if necessary, assume $g(x_n) \geq \delta$. Pick $u \in S_X$ with $g(u) > 1 - \eta$, where $\eta > 0$ is small. For any $\lambda > 1$, sequential ASQ gives $\|\lambda u + x_n\| \rightarrow \lambda$, while along the chosen subsequence

$$\|\lambda u + x_n\| \geq g(\lambda u + x_n) \geq \lambda(1 - \eta) + \delta.$$

Taking $\eta < \delta/(2\lambda)$ contradicts convergence to λ . Thus $g(x_n) \rightarrow 0$, as claimed. $\square$

Lemma 2. *Every real sequentially ASQ Banach space contains a complemented subspace isomorphic to c_0 .*

Proof. Let $(x_n) \subset S_X$ witness sequential ASQ, and choose $f_n \in S_{X^*}$ with $f_n(x_n) = 1$. By Lemma 1, $f_n(x) \rightarrow 0$ for each fixed x , and $x_n \rightarrow 0$ weakly.

We pass to a subsequence, still denoted (u_k) , with associated functionals (g_k) , satisfying two standard diagonal requirements. First, using the finite-dimensional selection lemma for sequentially ASQ sequences from [1, Lemma 3.2], we make (u_k) 2-equivalent from above to the c_0 -basis:

$$\left\| \sum_{k=1}^m a_k u_k \right\| \leq 2 \max_{1 \leq k \leq m} |a_k| \quad (m \in \mathbb{N}).$$

Second, using $g_n(x) \rightarrow 0$ for fixed x and weak nullity of (x_n) , we diagonalize further so that, for all $i \neq j$,

$$|g_i(u_j)| < 2^{-i-j-3}.$$

Indeed, at stage k only finitely many previous vectors and previous functionals are involved, so the two pointwise convergence statements allow choosing the next index large enough.

Let $Y = \overline{\text{span}}\{u_k : k \in \mathbb{N}\}$. Define

$$T : X \longrightarrow c_0, \quad Tx = (g_k(x))_{k=1}^\infty.$$

Lemma 1 gives $Tx \in c_0$ for every x , and $\|T\| \leq 1$. Also define $U : c_0 \rightarrow Y$ by

$$U(a) = \sum_{k=1}^\infty a_k u_k.$$

The upper c_0 -estimate above makes U bounded.

Consider $A = TU : c_0 \rightarrow c_0$. Its matrix is $A_{ij} = g_i(u_j)$. The diagonal entries are 1, and the off-diagonal estimate gives

$$\|A - I_{c_0}\| \leq \sup_i \sum_{j \neq i} |g_i(u_j)| < \frac{1}{2}.$$

Hence A is invertible on c_0 . It follows that U is an isomorphic embedding of c_0 into X , and

$$P = UA^{-1}T : X \longrightarrow Y$$

is a bounded projection, since $PU = UA^{-1}TU = U$. Therefore Y is a complemented copy of c_0 in X . $\square$

Counterexample

Theorem 1. *The real Banach space ℓ_∞ contains an isomorphic copy of c_0 but does not admit any equivalent sequentially ASQ norm.*

Proof. The canonical inclusion $c_0 \subset \ell_\infty$ shows that ℓ_∞ contains c_0 . Suppose that an equivalent norm $|\cdot|$ on ℓ_∞ were sequentially ASQ. By Lemma 2, the renormed space $(\ell_\infty, |\cdot|)$ would contain a complemented subspace isomorphic to c_0 . Complementability of a closed subspace is preserved under equivalent renorming of the ambient Banach space, so the usual ℓ_∞ would contain a complemented copy of c_0 .

This contradicts Phillips’ classical theorem, in the standard form that every bounded operator $\ell_\infty \rightarrow c_0$ is weakly compact [2]. Indeed, a complemented copy $Y \subset \ell_\infty$ with $Y \simeq c_0$ would give an onto operator $\ell_\infty \rightarrow c_0$, obtained by composing the projection onto Y with an isomorphism $Y \rightarrow c_0$. By the open mapping theorem, an onto weakly compact operator would make the unit ball of c_0 relatively weakly compact, forcing c_0 to be reflexive. This is impossible. Hence no equivalent norm on ℓ_∞ is sequentially ASQ. $\square$

Consequence for the Source Question

The source paper’s question “whether every Banach space containing c_0 can be renormed to be sequentially ASQ” has a negative ZFC answer. The counterexample is ℓ_∞ . This does not address the separate set-theoretic question in [1] about removing selective ultrafilters from their fourth-dual theorem for already sequentially ASQ spaces.

Novelty and Search Bounds

The run indexes were searched for 2110.14313, sequentially ASQ, sequentially almost square, renormed to be sequentially ASQ, complemented c_0 , and ℓ_∞ . The only nearby packet records the known nonsequential ASQ renorming theorem for spaces containing c_0 . Web searches on 2026-06-20 for exact phrases around the source question and the obstruction above found the source paper but no later answer.

References

- [1] A. Aviles, G. Martinez-Cervantes, and A. Rueda Zoca, *Banach spaces containing c_0 and elements in the fourth dual*, arXiv:2110.14313, 2021.
- [2] R. S. Phillips, *On linear transformations*, Transactions of the American Mathematical Society **48** (1940), 516–541.